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Author(s): F. E. Spencer and A. V. Phelps

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MOMENTUM TRANSFER CROSS-SECTIONS AND CONDUCTIVITY INTEGRALS FOR GASES OF MHD INTEREST

by

F. E. Spencer, Jr.
Pittsburgh Energy Research Center
U. S. Energy Research and Development Administration
Pittsburgh, Pennsylvania 15213

A. V. Phelps
Joint Institute for Laboratory Astrophysics
University of Colorado and National Bureau of Standards
Boulder, Colorado 80309

Abstract

We give our present best estimates for the electron momentum-transfer collision cross-sections for the principal gases occurring in present laboratory and demonstration scale MHD generators of the closed cycle type and of those open cycle generators running on ash-free combustion plasmas. Also, we give the integrals of these cross-section functions for direct incorporation into the algorithm of Demetriades for calculating electrical properties of gas mixtures.

I. Introduction

Once the composition (64, 65, 66) of a combustion mixture is known, electrical properties (particularly scalar conductivity and electron mobility) must be calculated (1, 15, 22, 27, 28, 36, 38) in order to proceed with modeling calculations for MHD generators. We have attempted here to give one more-or-less complete set of electron momentum-transfer collision cross-sections for use in any of the currently popular methods for calculating these electrical parameters.

In Table 1 we give our recommended cross-sections Q_m (in 10^{-20} m^2) for the gases Ar, CO, CO_2 , Cs, CsOH , H, H_2 , H_2O , K, KOH, NO, N_2 , O, OH, O_2 , and SO_2 , as a function of incident electron energy ϵ (in eV). The data for the individual gases are discussed in separate sections below, after a brief discussion of our estimation scheme for molecular dipoles.

Much of the data is the same as that given earlier (55) for gases of atmospheric interest. It is worth noting that species of relatively low concentration, which have large cross-sections, may play a dominant role in determining the plasma conductivity. Unfortunately, some of these species have the least well determined cross-sections, such as H_2O and KOH.

Several reviews (16, 32, 34) of electron collision frequency data have appeared recently. By comparison with these, it will be noted that we often refer to only a limited number of the data sources. This is done for brevity, but we believe it is justified by the reliability of the data cited.

II. Estimating Cross-Sections for Molecular Dipoles

When the dipole moment for a molecule is known, the plane-wave Born approximation (3) gives an approximate value for the electron momentum-transfer collision cross-section from quantum

mechanical considerations. This approximation assumes that the interaction between the electron and the permanent dipole is the only significant process and that the cross section is dominated by contributions from collisions at large impact distances. Such calculations neglect vibrational and electronic transitions and rotational transitions other than $|\Delta J|=1$. Altschuler's formula (3) gives the cross-section as

$$Q_m = \frac{8\pi}{3} \left(\frac{2\pi D e}{h\nu} \right)^2 \quad (\text{electrostatic units})$$

where D is the dipole moment, h is Planck's constant and v is the electron speed.

More detailed work since that of Altschuler (3) has shown that his hypotheses are usually not justified. However, a log-log regression of known collision frequencies (at 300 kelvins) against dipole moments for thirty-four molecules (7) for which both measurements are available leaves little doubt that dipole moment is a primary variable in determining cross-section, when the moment is large. We are led to believe that 60% of the Altschuler value (3) is a reasonable estimate for highly polar molecules when better information is unavailable. We assume, at low energy, that the cross-section varies as the reciprocal of the electron energy, as in Altschuler's formula. For small dipole moments, the Altschuler values are usually low, so for molecular dipoles of moment less than about 2.5 debyes (e.g., OH and SO_2), we have scaled from the measured values of Q_m for water, using the square of the dipole moment ratio as proportionality constant.

Our tables which are estimated using these "theories" may be expected to change as more detail becomes available. Work is currently in progress at JILA (by Norcross and Collins) which may yield some changes. The user of our estimates is especially notified to be on the alert for them.

The problem of measuring dipole moments for MHD-relevant gases, becomes, in light of the above, a question of some interest. The moment for KOH, for example, has not been published, but measured values for LiOH (21) and CsOH (39) are available. As we will see in section XII, $D(\text{KOH})$ is estimated.

A related problem which should be mentioned in this context, is that of dipole moment dependence upon the vibration state of the hydroxide. Freund (21) shows that the dipole moment for LiOH increases from 4.755 in the ground state to 5.133 (both, debyes) in the ($J=1$, 030)-state. Assuming the Altschuler dependence of cross-section upon dipole

Table 1. Recommended $Q_m(\epsilon) \times (10^{+20})m^2$

ϵ/eV	Ar	CO	CO ₂	Cs	CsOH	H	H ₂	H ₂ O
0.	7.5	60.0	600.	35000	149,000	39.6	6.4	50000.
.001	7.5	40.0	540.	30000	149,000	40.4	6.4	33000.
.002	7.1	25.0	380.		75,000	40.6	6.5	16500.
.003	6.7	17.7	307.		50,000	40.8	6.6	11000.
.005	6.1	14.0	237.		30,000	41.1	6.8	6600.
.007	5.4	9.8	200.		21,300	41.3	7.1	4710.
.0085	5.05	8.6	182.		17,500	41.4	7.2	3880.
.010	4.6	7.8	170.	10000	14,900	41.5	7.3	3300.
.015	3.75	6.5	138.	7100	9,900	41.6	7.7	2120.
.02	3.25	5.9	120.	5550	7,500	41.8	8.0	1610.
.03	2.5	5.4	97.	3530	5,000	41.84	8.5	1060.
.04	2.05	5.2	85.	2340	3,700	41.8	8.96	830.
.05	1.73	5.4	76.	1620	2,980	41.7	9.28	650.
.07	1.13	6.1	64.	800	2,130	41.3	9.85	456.
.10	.59	7.3	52.	305	1,490	40.6	10.5	318.
.12	.40	7.9	46.	307	1,240	39.7	10.85	265.
.15	.23	8.8	40.	650	990	38.8	11.4	210.
.17	.16	9.4	36.	1020	870	38.2	11.6	187.
.20	.103	10.0	31.5	1570	750	37.3	12.0	153.
.25	.091	11.2	25.	1793	600	36.0	12.5.	124.
.30	.153	12.0	20.	1490	500	35.0	13.	102.
.35	.235	12.7	16.5	790	425	33.7	13.45	89.
.40	.33	13.5	14.	458	373	32.7	13.9	78.
.50	.51	14.3	10.7	202	298	31.0	14.7	63.5
.70	.69	15.5	7.6	82	213	27.4	16.3	46.3
1.0	1.38	17.0	5.8	80	149	25.2	17.4	33.1
1.2	1.37	23.0	5.3	114	124	23.8	17.8	28.
1.3	1.82	31.0	5.1	134	115	23.0	18.0	26.
1.5	2.1	35.0	5.05	156	99.3	21.7	18.25	22.9
1.7	2.32	30.0	5.1	161	87.7	20.5	18.25	20.
1.9	2.58	20.0	5.2	158	78.4	19.4	18.1	18.2
2.1	2.85	15.0	5.35	154	71.0	18.5	17.9	16.6
2.2	2.97	14.0	5.65	152	67.7	18.1	17.7	16.0
2.5	3.37	12.6	6.5	146	59.6	16.8	17.2	14.4
2.8	3.8	12.1	7.5	141	53.2	15.8	16.7	13.2
3.0	4.1	12.0	8.9	137	49.7	15.2	16.1	12.4
3.3	4.5	11.8	11.1	132	45.1	14.3	15.6	11.6
3.6	4.9	11.7	14.3	129	41.4	13.4	15.0	10.8
4.0	5.4	11.7	15.3	125	37.3	12.4	14.3	10.0
4.5	6.1	11.7	14.8	121	33.1	11.3	13.6	9.3
5.0	6.7	11.6	13.	105	29.8	10.2	12.9	8.6
6.0	8.1	11.5	10.3	100	24.8	8.4	11.8	7.55
7.0	9.6	11.4	9.8	95	21.3	7.1	11.0	7.05
8.0	11.7	11.2	10.8	85	18.6	6.1	10.3	6.7
10.0	15.0	11.0	12.		14.9		9.0	6.6
12.0	15.2	11.3	13.2		12.4		8.1	6.65
15.0	14.1	11.9	14.5		10.3		7.1	7.4
17.0	13.1	12.2	14.9		9.4		6.6	7.9
20.0	11.5	12.6	15.6		8.2		5.9	8.4
25.0	9.0	12.2	16.0		7.0		5.1	8.6
30.0	7.2	11.6	15.9		6.3		4.55	8.3
50.0	4.8	9.8	12.9		4.7		3.4	5.0
75.0	3.5	8.4	9.5		3.8		2.3	4.1
100.0	2.8	7.4	8.0		3.3		1.8	3.5

Table 1. (continued) Recommended $Q_m(\epsilon) \times (10^{+20})_m^2$

ϵ/eV	K	KOH	NO	N ₂	O	OH	O ₂	SO ₂
0.	676	180,000	120	1.10		40000	.35	39000
.001	350	180,000	94	1.36		26400	.35	25700
.002	243	90,000	58	1.49		13200	.36	12900
.003	189	60,000	38	1.62		8800	.40	8600
.005	200	36,000	22.5	1.81		5280	.50	5150
.007	433	25,700	16.3	2.0		3800	.58	3670
.0085	680	21,200	13.3	2.1		3100	.64	3030
.010	1776	18,000	12.	2.19		2640	.70	2570
.015	7200	12,000	8.1	2.55		1700	.87	1650
.02	4300	9,000	6.2	2.85		1300	.99	1250
.03	1840	6,000	4.65	3.4		850	1.24	830
.04	1220	4,500	3.85	3.85		660	1.44	650
.05	960	3,600	3.4	4.33		520	1.60	510
.07	720	2,570	3.0	5.1		360	2.1	360
.10	550	1,800	3.1	5.95	4.8	257	4.2	250
.12	500	1,500	3.45	6.45	4.82	214	4.38	206
.15	435	1,200	4.05	7.1	4.85	170	4.65	164
.17	405	1,060	4.55	7.4	4.87	151	4.8	146
.20	360	900	5.3	7.9	4.9	122	5.1	120
.25	313	720	7.0	8.5	4.92	100	5.3	97
.30	270	600	11.1	9.0	4.95	82	5.46	80
.35	243	514	17.5	9.4	4.97	71	5.65	69
.40	215	450	22.5	9.7	5.0	62	5.8	61
.50	180	360	29.	9.9	5.05	51	5.9	50
.70	136	257	22.5	10.0	5.15	37	6.1	36
1.0	96	180	16.1	10.0	5.2	26	7.2	26
1.2	81	150	14.5	10.4	5.25	22	7.9	22
1.3	76	138	13.7	11.0	5.3	21	7.9	20.2
1.5	66	120	12.9	12.0	5.35	18.3	7.6	17.8
1.7	51	106	11.7	13.8	5.4	16.1	7.4	15.6
1.9	67	94.7	11.1	19.6	5.43	14.5	6.9	14.1
2.1	66	85.7	10.7	27.0	5.46	13.2	6.6	12.9
2.2	63	81.8	10.4	28.5	5.49	12.8	6.5	12.4
2.5	54	72.0	9.9	30.0	5.6	11.5	6.1	11.2
2.8	47	64.3	9.3	28.0	5.65	10.6	5.8	10.3
3.0	43	60.0	9.1	21.7	5.7	9.9	5.7	9.7
3.3	39.5	54.5	8.6	17.2	5.75		5.5	
3.6	36.5	50.0	8.3	14.7	5.85		5.45	
4.0	33.5	45.0	8.1	12.6	6.0		5.5	
4.5	32	40.0	8.1	11.3	6.1		5.55	
5.0	31	36.0	8.1	10.9	6.2		5.6	
6.0		30.0	8.2	10.4	6.4		6.0	
7.0		25.7	8.3	10.1	6.6		6.6	
8.0		22.5	8.4	10.0	6.75		7.1	
10.0		18.0	8.8	10.2	7.2		8.0	
12.0		15.0	9.9	10.4	7.45		8.5	
15.0		12.6	11.5	11.1	7.8		8.8	
17.0		11.3	11.4	11.6	7.7		8.7	
20.0		9.7	11.0	12.0	7.6		8.6	
25.0		8.3	10.7	11.7	7.2		8.2	
30.0		7.2	10.3	11.1	6.6		8.0	
50.0		5.3	9.2	9.3	6.3		7.7	
75.0		4.6	8.2	7.7	6.1		6.8	
100.0		3.7	7.5	6.6	6.0		6.5	

moment, this means that LiOH may have an increased cross-section, by about 16.5%, in its excited states. This percent change is well within our estimation tolerances at this time, but may lead to interesting results when really precise data become available: that is, cross sections for the triatomic molecules may be dependent on the gas temperature. In the contrary direction, Maki (43) shows that some dipolar triatomics (viz.: OCS) have decreased dipole moments in their excited states.

III. Argon (Ar)

The momentum transfer collision cross sections for electrons in argon have been and still are the subject of considerable controversy (14, 23, 26). The recommended values for electron energies below 9 eV given in Table 1 are from Crompton et al. (14). At energies above 9 eV we have given the results of Fletcher and Burch (19).

IV. Carbon Monoxide (CO)

The momentum transfer cross sections given for energies below 1 eV in Table 1 are from Hake and Phelps (29) and agree well with theoretical predictions (12). Momentum transfer cross sections at higher energies have been calculated by Takayanagi (69) for a few energies from elastic differential cross-section data and, together with total cross-section data (55), have been used (29) to fill out Table 1. Individual values in this energy range could easily be in error by $\pm 40\%$.

V. Carbon Dioxide (CO₂)

The momentum transfer cross sections listed for electrons in CO₂ in Table 1 are from Lowke et al. (41). The values differ somewhat from those of Hake and Phelps (29) primarily because of differences in the inelastic collision cross sections used in the analyses of electron transport data. Although there is still not sufficient data to assign a unique set of inelastic (vibrational excitation) cross sections (4) more recent analyses (57) have not suggested significant changes in the momentum transfer cross sections.

VI. Cesium (Cs)

Nighan and Postma (50) analyzed swarm data from three sources (8, 52, 59). The tabulated numbers are a digitization of their reported curve. Below .02 eV, their data are extrapolated.

VII. Cesium Hydroxide (CsOH)

No swarm data or theoretical calculations are available. Below 10 eV electron energy we estimate the cross section as 60 ± 20 percent of the values given by the Born approximation (3), using the measured (39) dipole moment of 7.1 ± 0.5 debyes. Note that the dipole moment for CsOH is given in terms of measured values (49) for CsF and OH by $D(\text{CsF}) - D(\text{OH})/2$ rather than $D(\text{CsF}) - D(\text{OH})$ as suggested by Krauss (37). The uncertainty assigned to this cross section is large enough to include values of the effective dipole moment obtained by interpolating between values obtained from mobility measurements (7) and the value obtained by fitting the calculation of Allison (2) to the energy dependence given by the Born approximation (3). Obviously, theory or experiments which reduce this uncertainty are crucial to an evaluation of the

role of CsOH in combustion MHD.

At electron energies above 10 eV we have assumed the cross section to have values typical of other molecules in Table 1.

VIII. Monatomic Hydrogen (H)

Momentum transfer collision cross sections for electrons in H have been calculated from theory and experiment by Luft and Kieffer (42) and are listed in Table 1.

IX. Hydrogen (H₂)

The momentum transfer collision cross sections given in Table 1 for H₂ for electron energies below 2 eV are from Crompton, Gibson, and McIntosh (13). At electron energies above 2 eV the values listed are from Engelhardt and Phelps (17) since other data (20) is incomplete. At these higher electron energies it has long been recognized (20) that significant changes in $Q(\epsilon)$ may be necessary when the electron transport data for H₂ is re-analyzed taking into account more recent transport and scattering results.

X. Water (H₂O)

The very large cross sections for momentum transfer collisions for low energy electrons in H₂O listed in Table 1 are from Cohen and Phelps (9). These values are based on electron transport data available in 1966 (54). Very recent electron transport data (73) show significant deviations from calculated values (9), so that the higher energy momentum transfer cross section values should be used with some caution. One source of concern is the validity of the spherical harmonic expansion technique for the solution of the Boltzmann equation used to analyze the nonthermal electron transport data for electrons in gases, such as H₂O, in which inelastic (rotational excitation) collision cross sections exceed elastic cross sections (61).

XI. Potassium (K)

Between 0 and 5 eV, our selected data for potassium is based on recent, but preliminary, detailed quantum mechanical calculations (53) and private communications, Norcross 1976). It appears that available determinations (31) of the cross sections are measurements of electrical conductivity over a limited temperature range in which the electron density is not measured directly. Below 1.0 eV the calculations are by Norcross--above 1.0 eV by Moores. Their calculations agree between 0.5 and 1.0 eV. Between 0.2 eV and 2.0 eV, theory (33) agrees with experimental differential cross sections (25, 72).

XII. Potassium Hydroxide (KOH)

Swarm data, dipole moment data and direct cross-section calculations are unavailable, so we resort to the "theory" of section II. From measurements (21, 39, 49), we have: $D(\text{CsOH}) = 7.1 \pm 0.5$; $D(\text{CsF}) = 7.88$; $D(\text{OH}) = 1.66$; $D(\text{LiF}) = 6.33$; $D(\text{LiOH}) = 4.76$; $D(\text{KF}) = 8.60$, all given in debyes. We are led to estimate $D(\text{KOH}) = D(\text{KF}) - 0.5$ $D(\text{OH}) = 7.77$ debyes, with an uncertainty of 0.7 debyes. Below 10 eV, the cross-sections in Table 1 are estimated as 60 ± 20 percent of the Born

approximation value (3). The uncertainty assigned to this cross section is large enough to include values of the effective dipole moment obtained by interpolating between values obtained from mobility measurements (7) and the value obtained by fitting the calculation of Allison (2) to the energy dependence given by the Born approximation (3). Obviously, theory or experiments which reduce this uncertainty are crucial to an evaluation of the role of KOH in coal-fired MHD.

At electron energies above 10 eV we have assumed the cross section to have values typical of other molecules in Table 1.

XIII. Nitric Oxide (NO)

The momentum transfer collision cross sections for NO given in Table 1 are from an analysis by Cohen and Phelps (10). At electron energies below 0.02 eV the cross section is calculated using the Born approximation developed by Altschuler (3) and a permanent dipole moment of 0.15 Debye. At electron energies between about 0.1 and 2 eV, electron transport data (6) was used. Note that these cross sections are as much as five times those derived from microwave measurements (45).

XIV. Nitrogen (N₂)

The momentum-transfer cross section data (18) tabulated for N₂ are expected to be accurate to about ± 10 percent for electron energies between 0.003 and about 4 eV and for vibrational temperatures below 500 kelvins. For electron energies corresponding to temperatures between 150 and 2000 kelvins the momentum-transfer collision frequency is reasonably well represented by a linear dependence on electron energy (56). The effect of vibrationally excited states of N₂ on the cross sections expected at elevated gas temperatures has been assumed to be small, i.e., less than 20 percent for vibrational temperatures below about 5000 kelvins.

The question as to the role of vibrational excitation was raised by Cotterau and Valentin (11), who interpret apparent low electrical conductivities in shock heated (6000 kelvins) nitrogen in terms of a much larger momentum transfer cross section for vibrationally excited N₂ than for ground state N₂. While some change in the derivative of the total scattering cross section has been observed in vibrationally excited N₂ by Burrows (47) it seems very unlikely that the increase in the momentum transfer cross section is more than 20 percent. Also, Klepis and Louis (35) apparently failed to observe a change in the effective momentum transfer collision frequency in shock heated N₂ at 4500 kelvins.

At electron energies above about 4 eV there is disagreement as to the correct cross sections (and collision frequencies) for momentum transfer. Recent experimental measurements of differential scattering cross sections lead to cross sections significantly below (63) and in fair agreement (67) with those (18) of Table 1. It is to be noted that the cross sections of Table 1 include significant contributions from inelastic collisions at energies near 2 eV and above 10 eV.

XV. Monatomic Oxygen (O)

The cross sections given in Table 1 for atomic oxygen are based primarily on the "total" cross-section measured by electron-beam techniques (50, 68). These values are about the mean of the two microwave measurements of the momentum-transfer cross section at the lower electron energies (40). Unfortunately, there is no better agreement among theoretical calculations (70) than among experiments (40), especially at the lower electron energies of interest here.

XVI. Hydroxyl (OH)

Since there are no experimental values for $\sigma_m(\epsilon)$ for OH we must rely entirely on theory (3). The values listed in Table 1 for electron energies below 3 eV are obtained by multiplying the values for H₂O by the square of the ratio of the dipole moments (49), i.e., by $(1.66/1.8)^2 = 0.85$.

XVII. Oxygen (O₂)

The determination of momentum transfer collision cross sections for electrons in O₂ is difficult because of large rates of electron attachment. Recently (43) measurements of electron drift velocity have been made at low enough electric field to gas density ratios (E/N) so that the electrons should be in thermal equilibrium with the gas. Unfortunately, the concurrent measurements of the ratio of longitudinal diffusion to mobility do not make possible a determination of the electron energy scale for these measurements and so do not allow an accurate determination of the energy dependent momentum transfer cross section. An analysis of these data is currently underway (58) and suggests the somewhat larger and more rapidly varying momentum transfer cross section at electron energies near 0.1 eV given in Table 1 than that deduced from microwave measurements (45). At electron energies between about 1 and 10 eV there is no reason to change the values given by Hake and Phelps (30). Again we must keep in mind that the momentum transfer cross section of interest in calculations of electron mobility includes contributions of inelastic collisions (71). The effects of vibrational excitation on the effective momentum transfer cross section via the sharply peaked cross sections (62) are unknown but will be assumed to be small.

XVIII. Sulphur Dioxide (SO₂)

Since there appears to be no experimental electron transport data for SO₂ the values listed in Table 1 for energies below 3 eV are calculated from theory (3) by multiplying the values for H₂O by the square of the ratio of the dipole moments (48), i.e., by $(1.63/1.85)^2 = 0.78$.

XIX. Integration of Parameters

We have, due to the popularity of the method, integrated these cross-section data to yield the interaction parameters necessary to apply the algorithm of Demetriades and Argyropoulos (15) to them when a composition is known. Please see the next section, where errors in mobilities calculated by this method are cited.

The parameters calculated are, in order:

$$\bar{Q}(r) \equiv \frac{1}{(r+1)!} \int_0^\infty e^{-x} x^{r+1} Q_m(x) dx$$

for $r = 1, 2, 3$;

$$C \equiv \bar{Q}(2)/\bar{Q}(1);$$

$$B \equiv (5\bar{Q}(2) - 4\bar{Q}(3))/\bar{Q}(1);$$

$$QDA \equiv \bar{Q}(1);$$

$$A2 \equiv 1.2C - 1;$$

$$A5 \equiv 2.5 - 1.2B;$$

where the integration variable is the reduced electron energy $x \equiv \frac{1}{2} mv^2/kT$, and the functions $Q_m(x)$ are obtained as linear splines from Table 1. Compare (5). Except for the linear spline fit, minimal numerical errors occur since on each reduced energy interval $a \leq x \leq b$, the integrals

$$\int_a^b x^j e^{-x} dx \quad \text{for } j = 0, \dots, 5$$

are all elementary and can be integrated exactly (to machine precision) by the parts formula. If the data ever were to justify, say, an escalation to cubic spline fitting, the integrals would remain elementary with j ranging up to 7 rather than 5. In cases where the tail of the cross-section function behaves as $1/x$, the integrals are still of the above form with $j = 0, \dots, 4$. For our present purposes this last observation has negligible numerical impact.

Tables of $QDA/(10^{-20} M^2)$, $A2$, and $A5$ as a function of electron temperature (T /kelvin) are given as an appendix.

In 1970, Argyropoulos and Casteel (5) gave similar computations for a few of the important gases in MHD plasmas based on the then current tabulation by one of us (AVP). Maxwell, et al., (44) gave computations for K and Cs. To the extent that our species list coincides with these two sources, we give a (historical) comparison of $QDA(2500)$ in Table 2. Neither Argyropoulos nor Maxwell estimated cross sections for $CsOH$, H , H_2 , KOH , O , OH , or SO_2 . Our value for $QDA(2900)$ for Cs is 19.1% below that of (44), and our value for $QDA(2900)$ for K is 39.1% below that of (44).

Table 2. - Historical Comparison of $QDA(2500)$'s

Species	(44)	(5)	This Work	$\Delta\%$
Ar		.568	.733	+29.0
CO		15.5	15.521	+ 0.1
CO ₂		11.1	12.340	+11.2
Cs ²	423		429.424	+ 1.5
H ₂ O		73.0	74.441	+ 2.0
K ²	292		190.746	-34.7
NO		20.6	18.437	-10.5
N ₂		9.82	9.818	- 0.0
O ₂		5.83	6.129	+ 5.1

XX. Comments on Application of Data

Some advantage may obtain from use of these data in an algebraic transport property algorithm such as was developed by Demetriades and Argyropoulos (15) in a manner related to the ideas of Grad (27, 28) and Kolodner (36), rather than the more widely accepted methods related to the ideas of Allis (1), Frost (22), and Kruger, et al. (38). Such advantages do not include great accuracy or overall computer speed.

Kruger, et al. (38) have shown that computations for electrons in Argon may be seriously in error if treated by a Chapman-Enskog type of approach unless it is of very high order. On the other hand, Mitchner relates in a private communication that calculations for combustion mixtures may yield usable results even for a 1st order Chapman-Enskog approach. Using the cross-sections of the present work, J. Land of JILA has calculated that the algorithm of Demetriades (15) yields mobilities 34% low for a pure Cesium gas at 2500 kelvins and 30-40% low for Argon. Further work on this problem is being pursued.

While a computer procedure for calculating electrical transport properties which entails numerical quadrature (1, 22, 38) is necessarily slower than a procedure based on an algebraic approach (15), the advantage disappears when one considers the large amount of time required for calculating the composition (64, 66) for a combustion mixture. We estimate that quadrature requires about 1% as much computer time as is required to solve the composition problem.

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AR				CO				CO2				CS			
T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5
300	1.356	-.2184	0.664	300	6.608	.3199	1.718	300	68.806	-.0373	0.851	300	1524.619	-.2164	1.206
1000	0.368	.0723	2.324	1000	10.625	.3585	1.643	1000	27.433	-.1430	0.892	1000	1001.231	.0495	0.537
1100	0.362	.1927	2.556	1100	11.029	.3552	1.639	1100	25.275	-.1445	0.891	1100	960.188	-.0011	0.460
1200	0.366	.3011	2.736	1200	11.405	.3529	1.643	1200	23.447	-.1460	0.890	1200	914.399	-.0481	0.419
1300	0.377	.3934	2.868	1300	11.758	.3518	1.658	1300	21.875	-.1474	0.889	1300	866.379	-.0908	0.404
1400	0.394	.4687	2.959	1400	12.094	.3523	1.680	1400	20.509	-.1488	0.888	1400	818.006	-.1291	0.408
1500	0.414	.5285	3.019	1500	12.417	.3543	1.708	1500	19.310	-.1500	0.889	1500	770.585	-.1631	0.426
1600	0.439	.5751	3.053	1600	12.734	.3576	1.740	1600	18.248	-.1510	0.891	1600	724.974	-.1929	0.454
1700	0.466	.6107	3.069	1700	13.045	.3620	1.771	1700	17.301	-.1518	0.895	1700	681.699	-.2189	0.490
1800	0.495	.6376	3.072	1800	13.356	.3671	1.800	1800	16.452	-.1523	0.901	1800	641.042	-.2414	0.531
1900	0.526	.6577	3.065	1900	13.666	.3725	1.824	1900	15.687	-.1524	0.907	1900	603.122	-.2605	0.576
2000	0.558	.6724	3.052	2000	13.976	.3779	1.841	2000	14.994	-.1521	0.915	2000	567.941	-.2765	0.624
2100	0.591	.6830	3.034	2100	14.287	.3830	1.850	2100	14.365	-.1514	0.925	2100	535.429	-.2897	0.673
2200	0.626	.6903	3.014	2200	14.598	.3874	1.851	2200	13.790	-.1503	0.935	2200	505.471	-.3001	0.723
2300	0.661	.6952	2.991	2300	14.908	.3909	1.844	2300	13.265	-.1487	0.947	2300	477.923	-.3081	0.773
2400	0.697	.6981	2.968	2400	15.217	.3935	1.829	2400	12.783	-.1467	0.960	2400	452.628	-.3138	0.823
2500	0.733	.6996	2.944	2500	15.521	.3949	1.807	2500	12.340	-.1443	0.974	2500	429.424	-.3173	0.872
2600	0.770	.6999	2.920	2600	15.821	.3952	1.779	2600	11.932	-.1414	0.990	2600	408.151	-.3188	0.920
2700	0.807	.6993	2.897	2700	16.115	.3944	1.746	2700	11.556	-.1380	1.007	2700	388.653	-.3185	0.966
2800	0.845	.6981	2.874	2800	16.401	.3925	1.709	2800	11.208	-.1342	1.025	2800	370.785	-.3166	1.010
2900	0.883	.6963	2.852	2900	16.678	.3895	1.668	2900	10.886	-.1299	1.044	2900	354.406	-.3131	1.052
3000	0.920	.6942	2.831	3000	16.946	.3856	1.626	3000	10.588	-.1251	1.064	3000	339.389	-.3083	1.092
3100	0.958	.6918	2.811	3100	17.202	.3808	1.582	3100	10.312	-.1199	1.086	3100	325.614	-.3023	1.130
3200	0.996	.6893	2.792	3200	17.447	.3752	1.537	3200	10.055	-.1141	1.110	3200	312.972	-.2953	1.165
3300	1.034	.6865	2.774	3300	17.680	.3689	1.492	3300	9.818	-.1079	1.134	3300	301.363	-.2873	1.197
3400	1.073	.6838	2.757	3400	17.899	.3621	1.448	3400	9.597	-.1012	1.160	3400	290.694	-.2785	1.227
3500	1.111	.6809	2.741	3500	18.106	.3547	1.405	3500	9.392	-.0941	1.187	3500	280.883	-.2690	1.255
3600	1.149	.6781	2.726	3600	18.300	.3470	1.363	3600	9.202	-.0864	1.215	3600	271.853	-.2589	1.280
4000	1.301	.6672	2.675	4000	18.941	.3133	1.210	4000	8.571	-.0515	1.338	4000	242.248	-.2151	1.356
4500	1.490	.6550	2.627	4500	19.460	.2696	1.062	4500	8.017	.0007	1.504	4500	216.153	-.1582	1.406
5000	1.678	.6448	2.593	5000	19.712	.2279	0.961	5000	7.664	.0590	1.667	5000	198.069	-.1053	1.417
5500	1.864	.6366	2.568	5500	19.755	.1906	0.898	5500	7.463	.1190	1.807	5500	185.185	-.0599	1.404
6000	2.048	.6300	2.551	6000	19.644	.1582	0.863	6000	7.379	.1763	1.912	6000	175.733	-.0227	1.376
8000	2.773	.6152	2.530	8000	18.453	.0749	0.879	8000	7.715	.3332	1.977	8000	154.754	.0575	1.223
10000	3.492	.6115	2.542	10000	17.034	.0432	0.972	10000	8.441	.3765	1.766	10000	143.826	.0742	1.089
12000	4.212	.6111	2.538	12000	15.831	.0384	1.061	12000	9.136	.3659	1.570	12000	135.637	.0659	0.991
14000	4.932	.6079	2.492	14000	14.893	.0462	1.131	14000	9.696	.3412	1.456	14000	128.413	.0488	0.922
16000	5.645	.5992	2.402	16000	14.177	.0593	1.183	16000	10.124	.3181	1.408	16000	121.707	.0293	0.875
18000	6.336	.5845	2.283	18000	13.630	.0741	1.224	18000	10.454	.3009	1.401	18000	115.419	.0101	0.842
20000	6.994	.5650	2.148	20000	13.211	.0891	1.256	20000	10.719	.2898	1.416	20000	109.529	-.0079	0.821

K				KOH				NO				N2			
T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5
300	1109.139	-.1944	0.958	300	3508.985	-.2077	0.901	300	5.798	.0053	1.323	300	5.272	.3849	1.680
1000	392.627	-.1087	0.937	1000	1025.266	-.2253	0.836	1000	10.569	.6582	2.473	1000	8.116	.3060	1.471
1100	364.833	-.1077	0.926	1100	925.762	-.2316	0.838	1100	11.746	.6264	2.216	1100	8.317	.2992	1.451
1200	341.217	-.1077	0.917	1200	842.308	-.2369	0.846	1200	12.836	.5888	1.986	1200	8.492	.2929	1.434
1300	320.824	-.1083	0.909	1300	771.456	-.2411	0.858	1300	13.821	.5492	1.789	1300	8.647	.2871	1.422
1400	302.981	-.1095	0.902	1400	710.707	-.2441	0.872	1400	14.692	.5101	1.622	1400	8.783	.2820	1.415
1500	287.198	-.1109	0.896	1500	658.192	-.2458	0.888	1500	15.449	.4725	1.482	1500	8.905	.2777	1.413
1600	273.109	-.1126	0.890	1600	612.472	-.2464	0.904	1600	16.098	.4371	1.366	1600	9.014	.2743	1.417
1700	260.434	-.1145	0.885	1700	572.413	-.2460	0.920	1700	16.645	.4040	1.270	1700	9.115	.2720	1.427
1800	248.953	-.1165	0.880	1800	537.110	-.2448	0.934	1800	17.100	.3734	1.191	1800	9.208	.2708	1.443
1900	238.493	-.1187	0.875	1900	505.831	-.2429	0.947	1900	17.471	.3450	1.126	1900	9.297	.2708	1.466
2000	228.911	-.1209	0.869	2000	477.976	-.2404	0.958	2000	17.769	.3189	1.072	2000	9.382	.2721	1.495
2100	220.092	-.1233	0.864	2100	453.055	-.2376	0.967	2100	18.002	.2949	1.029	2100	9.466	.2746	1.529
2200	211.940	-.1258	0.858	2200	430.653	-.2344	0.975	2200	18.179	.2728	0.993	2200	9.551	.2784	1.568
2300	204.375	-.1284	0.852	2300	410.432	-.2311	0.981	2300	18.306	.2525	0.963	2300	9.637	.2833	1.611
2400	197.329	-.1311	0.845	2400	392.105	-.2276	0.986	2400	18.390	.2338	0.940	2400	9.726	.2894	1.657
2500	190.746	-.1339	0.839	2500	375.426	-.2241	0.989	2500	18.437	.2165	0.921	2500	9.818	.2964	1.704
2600	184.576	-.1369	0.832	2600	360.199	-.2206	0.992	2600	18.452	.2007	0.905	2600	9.915	.3044	1.753
2700	178.776	-.1399	0.825	2700	346.242	-.2171	0.993	2700	18.441	.1860	0.893	2700	10.017	.3130	1.800
2800	173.310	-.1431	0.818	2800	333.408	-.2138	0.993	2800	18.406	.1725	0.884	2800	10.125	.3223	1.847
2900	168.147	-.1464	0.810	2900	321.568	-.2105	0.993	2900	18.352	.1601	0.877	2900	10.239	.3319	1.890
3000	163.259	-.1498	0.803	3000	310.613	-.2074	0.992	3000	18.281	.1486	0.872	3000	10.358	.3417	1.931
3100	158.622	-.1533	0.796	3100	300.448	-.2044	0.991	3100	18.196	.1380	0.869	3100	10.483	.3517	1.967
3200	154.215	-.1568	0.789	3200	290.994	-.2015	0.989	3200	18.099	.1281	0.866	3200	10.614	.3615	1.999
3300	150.019	-.1605	0.783	3300	282.169	-.1988	0.986	3300	17.993	.1190	0.865	3300	10.751	.3711	2.027
3400	146.017	-.1641	0.776	3400	273.922	-.1963	0.984	3400	17.879	.1106	0.865	3400	10.893	.3803	2.049
3500	142.195	-.1679	0.770	3500	266.192	-.1939	0.981	3500	17.758	.1028	0.866	3500	11.040	.3891	2.066
3600	138.539	-.1717	0.765	3600	258.929	-.1917	0.978	3600	17.633	.0955	0.867	3600	11.191	.3973	2.077
4000	125.368	-.1871	0.746	4000	233.787	-.1841	0.965	4000	17.098	.0714	0.877	4000	11.831	.4235	2.075
4500	111.552	-.2063	0.731	4500	208.995	-.1775	0.949	4500	16.408	.0498	0.895	4500	12.670	.4395	1.985
5000	99.994	-.2244	0.728	5000	189.327	-.1733	0.935	5000	15.739	.0349	0.916	5000	13.498	.4386	1.839
5500	90.199	-.2409	0.733	5500	173.274	-.1707	0.922	5500	15.113	.0248	0.937	5500	14.268	.4250	1.675
6000	81.819	-.2554	0.746	6000	159.868	-.1694	0.912	6000	14.537	.0183	0.958	6000	14.951	.4032	1.514
8000	58.116	-.2920	0.840	8000	122.581	-.1702	0.889	8000	12.715	.0135	1.041	8000	16.661	.2908	1.058
10000	44.032	-.2994	0.941	10000	99.606	-.1741	0.881	10000	11.493	.0266	1.118	10000	17.052	.1937	0.878
12000	35.130	-.2892	1.013	12000	83.917	-.1780	0.880	12000	10.666	.0474	1.190	12000	16.734	.1264	0.844
14000	29.190	-.2710	1.051	14000	72.492	-.1813	0.880	14000	10.102	.0714	1.257	14000	16.126	.0840	0.876
16000	25.025	-.2510	1.064	16000	63.803	-.1838	0.883	16000	9.717	.0963	1.319	16000	15.446	.0599	0.935
18000	21.973	-.2323	1.060	18000	56.970	-.1858	0.885	18000	9.458	.1208	1.374	18000	14.794	.0489	1.001
20000	19.650	-.2163	1.046	20000	51.463	-.1871	0.889	20000	9.291	.1441	1.421	20000	14.210	.0467	1.066

CSOH				H				H2				H2O			
T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5
300	2933.767	-.2022	0.898	300	40.873	.1810	1.234	300	9.807	.2711	1.463	300	635.354	-.2043	0.894
1000	874.258	-.2011	0.903	1000	36.257	.1362	1.137	1000	12.360	.2853	1.508	1000	185.862	-.2095	0.913
1100	794.597	-.2008	0.904	1100	35.691	.1318	1.129	1100	12.616	.2871	1.512	1100	168.649	-.2072	0.916
1200	728.280	-.2005	0.904	1200	35.150	.1278	1.122	1200	12.860	.2886	1.513	1200	154.379	-.2056	0.919
1300	672.216	-.2002	0.904	1300	34.632	.1240	1.116	1300	13.092	.2898	1.513	1300	142.366	-.2040	0.921
1400	624.196	-.1999	0.904	1400	34.138	.1206	1.111	1400	13.313	.2907	1.512	1400	132.118	-.2024	0.922
1500	582.605	-.1997	0.903	1500	33.664	.1175	1.107	1500	13.523	.2912	1.509	1500	123.276	-.2008	0.923
1600	546.231	-.1995	0.903	1600	33.211	.1146	1.103	1600	13.724	.2915	1.505	1600	115.570	-.1993	0.924
1700	514.149	-.1993	0.902	1700	32.778	.1120	1.100	1700	13.916	.2914	1.500	1700	108.794	-.1979	0.924
1800	485.638	-.1992	0.902	1800	32.362	.1095	1.097	1800	14.098	.2911	1.495	1800	102.791	-.1965	0.923
1900	460.132	-.1991	0.901	1900	31.964	.1072	1.094	1900	14.272	.2905	1.489	1900	97.435	-.1953	0.923
2000	437.178	-.1990	0.901	2000	31.581	.1051	1.091	2000	14.438	.2897	1.482	2000	92.626	-.1941	0.922
2100	416.411	-.1990	0.900	2100	31.214	.1031	1.088	2100	14.596	.2888	1.475	2100	88.284	-.1931	0.922
2200	397.529	-.1990	0.900	2200	30.861	.1012	1.086	2200	14.747	.2876	1.467	2200	84.344	-.1921	0.921
2300	380.285	-.1990	0.899	2300	30.521	.0993	1.083	2300	14.890	.2863	1.460	2300	80.752	-.1912	0.920
2400	364.477	-.1991	0.899	2400	30.193	.0976	1.080	2400	15.026	.2849	1.452	2400	77.464	-.1903	0.919
2500	349.927	-.1992	0.898	2500	29.876	.0959	1.077	2500	15.156	.2833	1.444	2500	74.441	-.1896	0.919
2600	336.496	-.1993	0.898	2600	29.570	.0942	1.074	2600	15.279	.2816	1.435	2600	71.654	-.1889	0.918
2700	324.054	-.1993	0.897	2700	29.274	.0926	1.071	2700	15.396	.2799	1.427	2700	69.075	-.1882	0.917
2800	312.499	-.1995	0.897	2800	28.988	.0910	1.067	2800	15.507	.2780	1.418	2800	66.681	-.1876	0.917
2900	301.735	-.1996	0.897	2900	28.710	.0894	1.064	2900	15.612	.2761	1.410	2900	64.453	-.1871	0.916
3000	291.686	-.1997	0.896	3000	28.440	.0879	1.061	3000	15.712	.2741	1.401	3000	62.374	-.1865	0.916
3100	282.281	-.1998	0.896	3100	28.178	.0864	1.058	3100	15.806	.2720	1.392	3100	60.430	-.1861	0.915
3200	273.465	-.1999	0.896	3200	27.924	.0848	1.055	3200	15.895	.2699	1.383	3200	58.608	-.1856	0.915
3300	265.176	-.2001	0.896	3300	27.676	.0833	1.051	3300	15.980	.2677	1.375	3300	56.895	-.1851	0.915
3400	257.376	-.2002	0.896	3400	27.434	.0819	1.048	3400	16.059	.2655	1.366	3400	55.285	-.1847	0.915
3500	250.619	-.2003	0.896	3500	27.199	.0804	1.045	3500	16.135	.2633	1.357	3500	53.765	-.1843	0.914
3600	243.067	-.2004	0.896	3600	26.969	.0789	1.042	3600	16.205	.2610	1.349	3600	52.330	-.1839	0.914
4000	218.723	-.2009	0.896	4000	26.103	.0731	1.029	4000	16.448	.2517	1.315	4000	47.306	-.1825	0.915
4500	194.354	-.2014	0.896	4500	25.120	.0661	1.014	4500	16.671	.2397	1.276	4500	42.278	-.1807	0.917
5000	174.848	-.2017	0.897	5000	24.228	.0592	1.000	5000	16.821	.2278	1.239	5000	38.254	-.1789	0.920
5500	158.883	-.2020	0.897	5500	23.410	.0525	0.986	5500	16.909	.2161	1.207	5500	34.959	-.1769	0.923
6000	145.570	-.2022	0.898	6000	22.654	.0459	0.973	6000	16.948	.2049	1.178	6000	32.214	-.1749	0.927
8000	108.991	-.2025	0.900	8000	20.102	.0209	0.926	8000	16.770	.1653	1.093	8000	24.679	-.1654	0.944
10000	87.076	-.2023	0.902	10000	18.076	-.0024	0.886	10000	16.306	.1343	1.044	10000	20.187	-.1543	0.963
12000	72.494	-.2019	0.902	12000	16.404	-.0238	0.855	12000	15.738	.1104	1.015	12000	17.222	-.1421	0.986
14000	62.044	-.2016	0.903	14000	14.993	-.0433	0.833	14000	15.149	.0917	0.996	14000	15.131	-.1285	1.013
16000	54.310	-.2011	0.903	16000	13.783	-.0606	0.818	16000	14.574	.0767	0.983	16000	13.592	-.1134	1.046
18000	48.265	-.2006	0.904	18000	12.736	-.0760	0.809	18000	14.029	.0644	0.974	18000	12.426	-.0966	1.084
20000	43.440	-.2000	0.906	20000	11.822	-.0894	0.805	20000	13.518	.0542	0.967	20000	11.517	-.0785	1.125

0				OH				02				SO2			
T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5	T/K	QDA	A2	A5
300	4.810	.2028	1.312	300	508.869	-.2095	0.903	300	2.652	.2276	2.044	300	497.601	-.2028	0.892
1000	4.917	.2117	1.331	1000	148.451	-.2074	0.912	1000	4.948	.3120	1.463	1000	145.665	-.2092	0.913
1100	4.931	.2124	1.332	1100	134.736	-.2062	0.915	1100	5.076	.3040	1.461	1100	132.154	-.2078	0.916
1200	4.944	.2130	1.333	1200	123.362	-.2048	0.917	1200	5.189	.2983	1.464	1200	120.955	-.2063	0.919
1300	4.958	.2136	1.334	1300	113.780	-.2033	0.919	1300	5.290	.2944	1.470	1300	111.529	-.2047	0.921
1400	4.970	.2140	1.335	1400	105.602	-.2019	0.920	1400	5.382	.2919	1.478	1400	103.489	-.2030	0.923
1500	4.983	.2145	1.336	1500	98.542	-.2005	0.920	1500	5.468	.2905	1.486	1500	96.552	-.2014	0.924
1600	4.994	.2149	1.336	1600	92.385	-.1992	0.921	1600	5.548	.2898	1.493	1600	90.508	-.1999	0.924
1700	5.006	.2152	1.337	1700	86.970	-.1979	0.921	1700	5.624	.2896	1.498	1700	85.195	-.1984	0.924
1800	5.017	.2155	1.337	1800	82.169	-.1968	0.920	1800	5.696	.2896	1.501	1800	80.468	-.1971	0.924
1900	5.027	.2159	1.338	1900	77.885	-.1957	0.920	1900	5.765	.2897	1.502	1900	76.208	-.1958	0.923
2000	5.038	.2162	1.339	2000	74.037	-.1947	0.919	2000	5.832	.2898	1.501	2000	72.518	-.1947	0.922
2100	5.048	.2164	1.340	2100	70.560	-.1937	0.919	2100	5.896	.2898	1.496	2100	69.114	-.1936	0.922
2200	5.058	.2167	1.340	2200	67.406	-.1929	0.918	2200	5.958	.2895	1.490	2200	66.026	-.1926	0.921
2300	5.067	.2170	1.341	2300	64.529	-.1921	0.918	2300	6.018	.2890	1.482	2300	63.210	-.1917	0.920
2400	5.076	.2173	1.342	2400	61.895	-.1913	0.917	2400	6.075	.2881	1.472	2400	60.633	-.1909	0.919
2500	5.085	.2176	1.343	2500	59.475	-.1906	0.917	2500	6.129	.2869	1.460	2500	58.264	-.1901	0.918
2600	5.094	.2179	1.344	2600	57.240	-.1900	0.917	2600	6.181	.2855	1.447	2600	56.079	-.1894	0.917
2700	5.103	.2181	1.345	2700	55.175	-.1894	0.916	2700	6.231	.2837	1.433	2700	54.058	-.1888	0.917
2800	5.111	.2184	1.346	2800	53.256	-.1888	0.916	2800	6.278	.2816	1.419	2800	52.181	-.1882	0.916
2900	5.120	.2187	1.347	2900	51.472	-.1883	0.916	2900	6.322	.2793	1.404	2900	50.435	-.1877	0.915
3000	5.128	.2190	1.348	3000	49.806	-.1878	0.915	3000	6.364	.2768	1.388	3000	48.806	-.1872	0.915
3100	5.136	.2193	1.349	3100	48.250	-.1873	0.915	3100	6.404	.2740	1.373	3100	47.282	-.1867	0.914
3200	5.144	.2196	1.350	3200	46.789	-.1868	0.915	3200	6.441	.2710	1.357	3200	45.854	-.1863	0.914
3300	5.152	.2200	1.351	3300	45.419	-.1863	0.915	3300	6.475	.2679	1.342	3300	44.512	-.1859	0.913
3400	5.159	.2203	1.352	3400	44.128	-.1859	0.915	3400	6.507	.2647	1.327	3400	43.249	-.1855	0.913
3500	5.167	.2206	1.354	3500	42.911	-.1855	0.915	3500	6.537	.2613	1.313	3500	42.058	-.1852	0.913
3600	5.175	.2209	1.355	3600	41.763	-.1851	0.915	3600	6.565	.2579	1.299	3600	40.933	-.1848	0.913
4000	5.204	.2223	1.360	4000	37.742	-.1836	0.915	4000	6.653	.2436	1.247	4000	36.993	-.1836	0.912
4500	5.240	.2241	1.366	4500	33.721	-.1818	0.916	4500	6.722	.2257	1.197	4500	33.048	-.1822	0.913
5000	5.274	.2259	1.373	5000	30.501	-.1802	0.916	5000	6.753	.2092	1.162	5000	29.889	-.1809	0.913
5500	5.308	.2279	1.379	5500	27.862	-.1788	0.916	5500	6.756	.1946	1.141	5500	27.299	-.1797	0.914
6000	5.342	.2298	1.386	6000	25.604	-.1775	0.916	6000	6.738	.1822	1.130	6000	25.137	-.1786	0.914
8000	5.472	.2376	1.409	8000	19.586	-.1742	0.910	8000	6.577	.1545	1.166	8000	19.169	-.1753	0.911
10000	5.599	.2446	1.427	10000	15.903	-.1731	0.902	10000	6.405	.1539	1.259	10000	15.556	-.1739	0.904
12000	5.722	.2506	1.441	12000	13.413	-.1738	0.895	12000	6.290	.1692	1.361	12000	13.118	-.1740	0.898
14000	5.840	.2557	1.452	14000	11.612	-.1750	0.891	14000	6.242	.1917	1.450	14000	11.354	-.1751	0.892
16000	5.954	.2599	1.459	16000	10.242	-.1768	0.887	16000	6.249	.2154	1.517	16000	10.015	-.1766	0.889
18000	6.063	.2633	1.463	18000	9.168	-.1782	0.886	18000	6.297	.2373	1.560	18000	8.963	-.1781	0.887
20000	6.167	.2659	1.463	20000	8.289	-.1807	0.883	20000	6.375	.2556	1.583	20000	8.111	-.1798	0.885